Free-fall lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate g from measured motion and discuss measurement limits.

SUCCESS CRITERIA - I can...

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “slope” and “trial” to explain your answer. Sentence starters are optional.

slope: vertical change divided by horizontal change on a graph

trial: one run or repetition of a test

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

Two equal **3000 kg** masses are **5 m** apart.
1. Use G=**6.67×10^{-11} N·m²/kg²**.
2. **Find force**.
3. By what factor is an answer wrong if r rather than r² is used?

Sentence starter: The step I would change is ... . I would change it because I found ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = \Delta v/\Delta t$; $v_f = v_i + a\,\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

Worked reasoning: Supplied trial values **8**, **10**, **12 m/s²** have mean **10 m/s²** and range **8**-**12**. The spread describes variation; averaging does not remove every systematic error.

Your turn: Plot the three values on a number line and **mark the mean**. **Name an error** that could shift all trials together.

Sentence starter: The step I would change is ... . I would change it because I found ... .

Trial	Estimated g (m/s²)
1	8
2	10
3	12

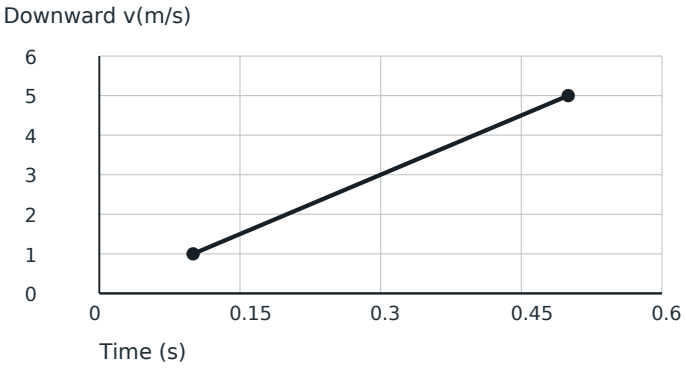
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. Use the supplied velocity-time graph to **estimate acceleration**.
2. **Show the change in velocity** and elapsed time.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

Three video trials estimate g as **9.6**, **9.9** and **10.2 m/s²**.
1. **Find the mean** and half-range.
2. **Compare the mean** with **9.8 m/s²** using percent difference from that reference.
3. **Label a sketch of the** measurement interval.

Word help: estimate = find an approximate value

Sentence starter: Both ... . They differ in ... because ... .

Trial	Estimated g (m/s²)
1	9.6
2	9.9
3	10.2

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Plan three repeated trials. **Name one part of the** setup you will keep the same, a measurement you will record and a reason not to include a badly pushed release without comment.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** lamp hangs at rest from a vertical rope. Use g=**9.8**. **Find tension**. Then **find tension** if it accelerates upward at **3 m/s²**.
Draw the forces on the chosen object and **label their sources**.

Word help: tension = the pulling force through a stretched rope or cable

Sentence starter: I found … [unit]. This value means … in this situation.

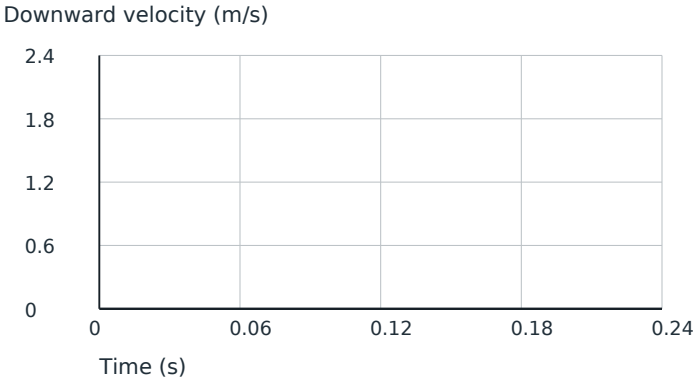
Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

A video gives downward velocities of **0.15 m/s** at **0.02 s** and **2.11 m/s** at **0.22 s**. Plot the points and **find the slope**. **Explain why fitting several reliable** measurements can be better than using one very short interval.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Supplied estimates are **9.4**, **9.8**, **10.2 m/s²**.
1. **Find mean**, half-range and percent deviation of mean from **9.81**.
2. Could zero spread guarantee accuracy?

Sentence starter: I found … [unit]. This value means … in this situation.

Group / trial	Values (m/s²)	Mean	Spread
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③